

A: Datasheet

Algorithm: kakao_002

Developer: Kakao Brain

Submission Date: 2023_11_07

Template size: 2048 bytes

Template time (2.5 percentile): 594 msec

Template time (median): 595 msec

Template time (97.5 percentile): 660 msec

Investigation:

Frontal mugshot ranking 77 (out of 449) -- FNIR(1600000, 0, 1) = 0.0013 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 70 (out of 411) -- FNIR(1600000, 0, 1) = 0.0088 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 10 (out of 380) -- FNIR(1600000, 0, 1) = 0.0559 vs. lowest 0.0517 from sensetime_009

Immigration visa-border ranking 8 (out of 338) -- FNIR(1600000, 0, 1) = 0.0008 vs. lowest 0.0006 from cloudwalk_mt_002

Immigration visa-kiosk ranking 4 (out of 282) -- FNIR(1600000, 0, 1) = 0.0419 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

Frontal mugshot ranking 26 (out of 449) -- FNIR(1600000, T, L+1) = 0.0021, FPIR=0.001000 vs. lowest 0.0011 from idemia_010

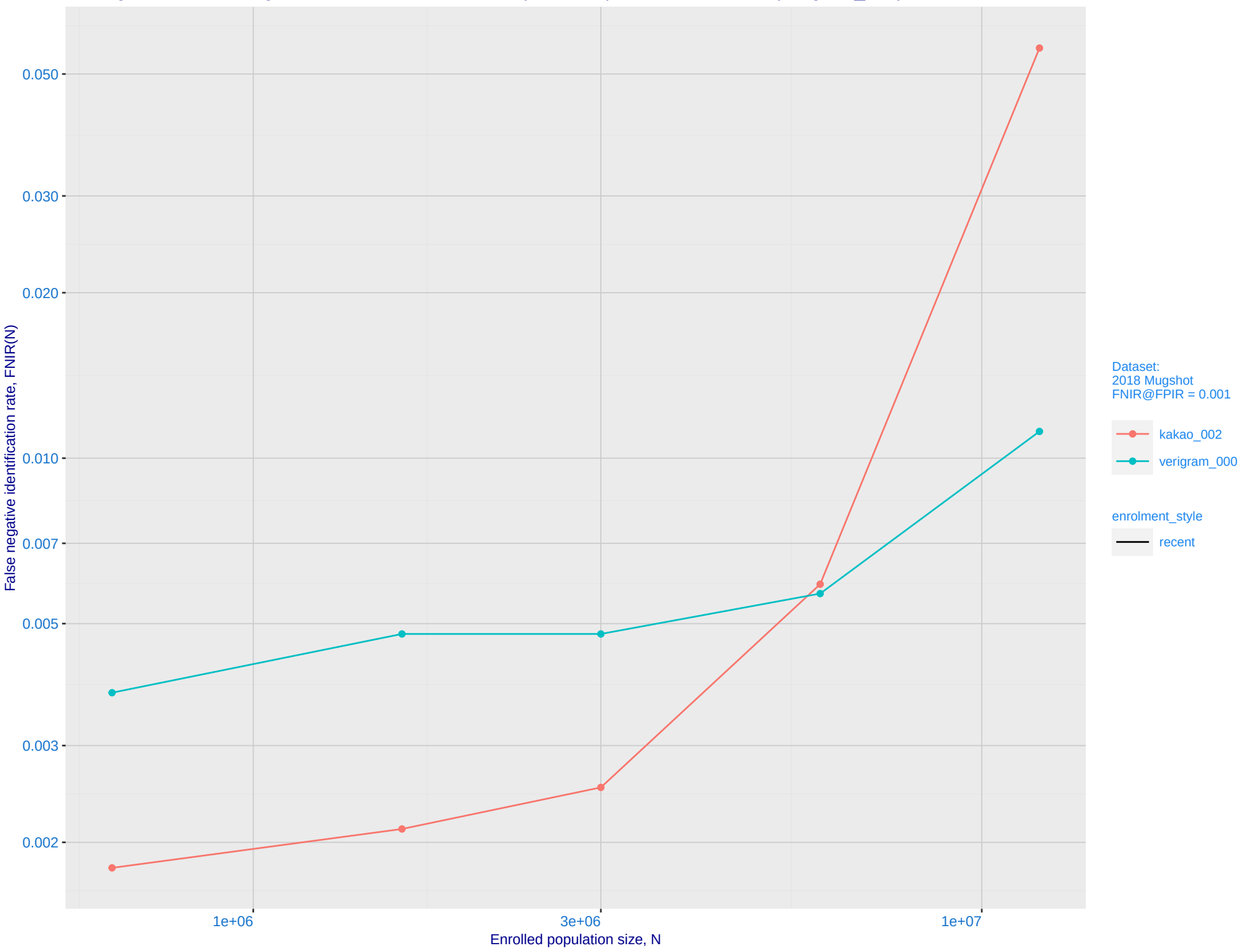
Mugshot webcam ranking 29 (out of 409) -- FNIR(1600000, T, L+1) = 0.0145, FPIR=0.001000 vs. lowest 0.0072 from sensetime_009

Mugshot profile ranking 10 (out of 379) -- FNIR(1600000, T, L+1) = 0.1437, FPIR=0.001000 vs. lowest 0.0634 from cloudwalk_mt_002

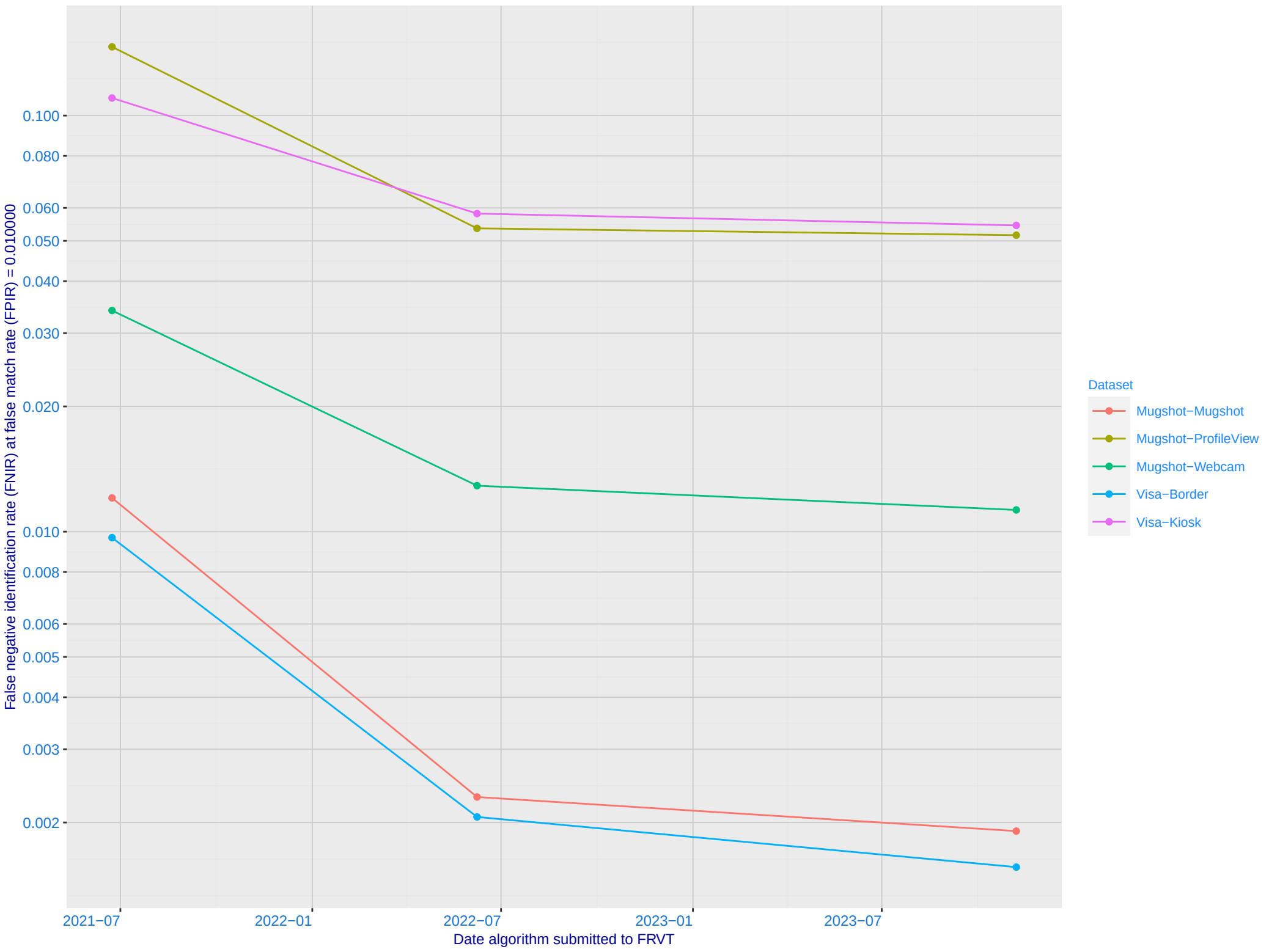
Immigration visa-border ranking 15 (out of 337) -- FNIR(1600000, T, L+1) = 0.0030, FPIR=0.001000 vs. lowest 0.0010 from cloudwalk_mt_002

Immigration visa-kiosk ranking 10 (out of 282) -- FNIR(1600000, T, L+1) = 0.0668, FPIR=0.001000 vs. lowest 0.0517 from cloudwalk_mt_002

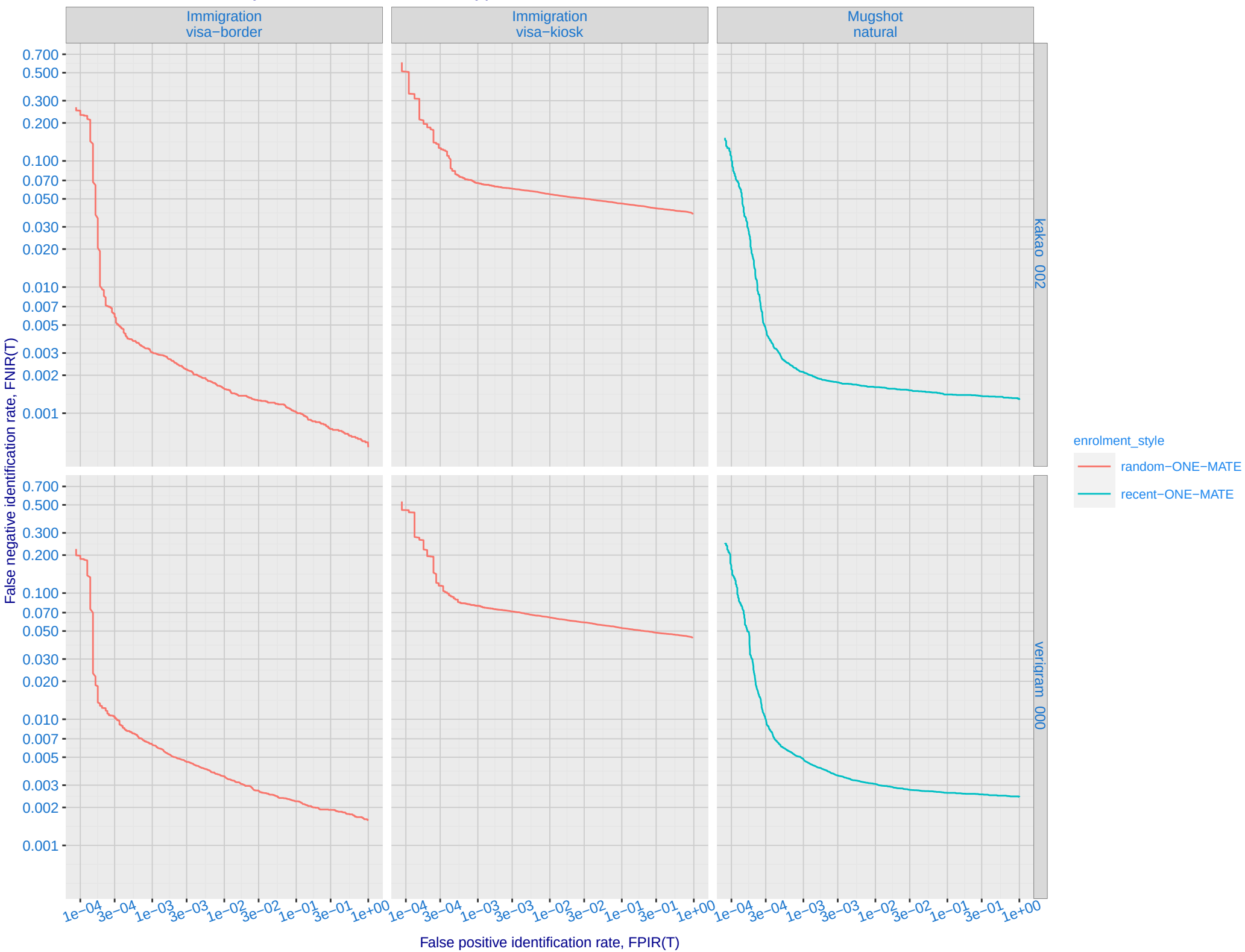
B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (verigram_000)



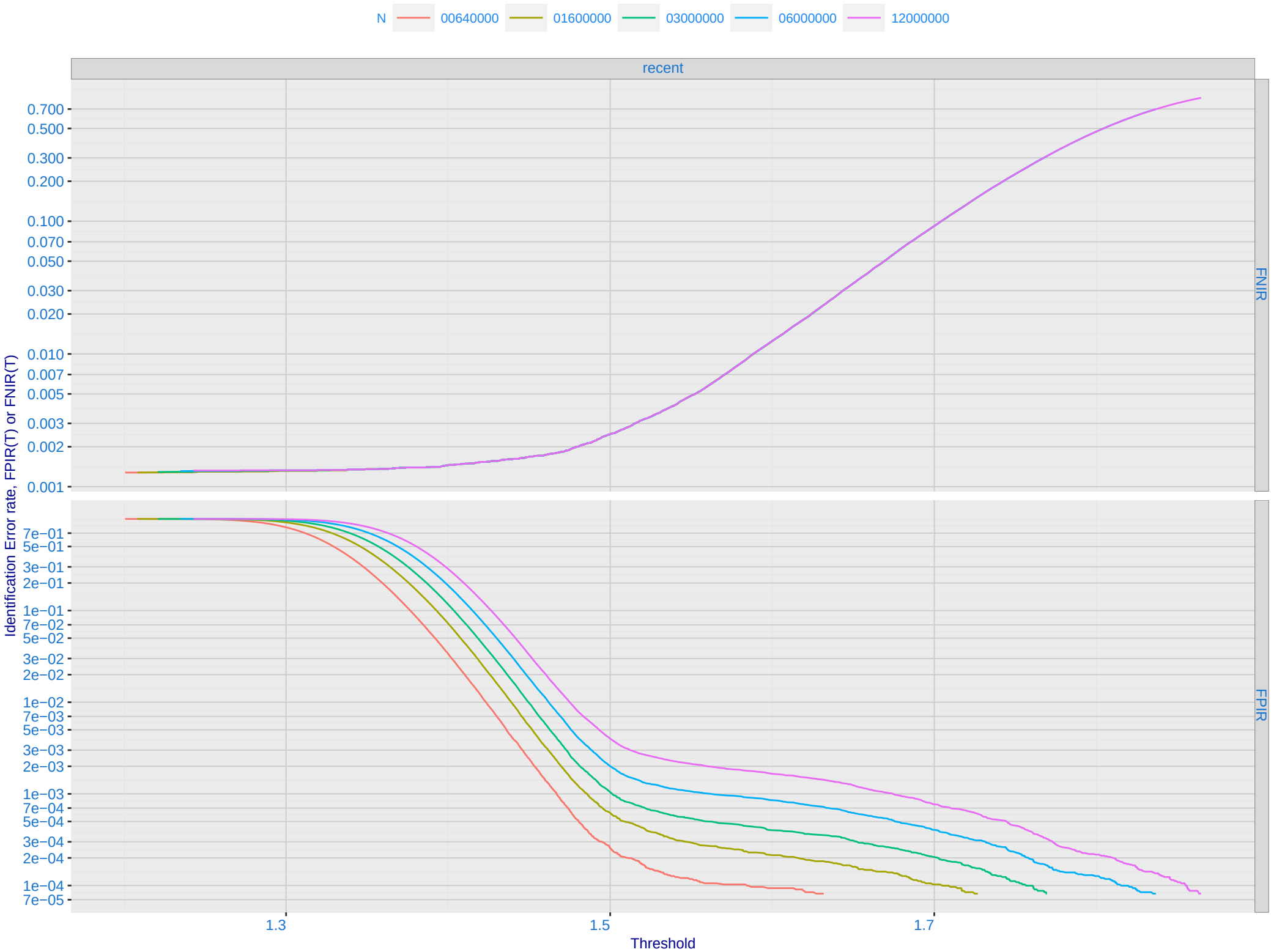
C: Evolution of accuracy for KAKAO algorithms on three datasets 2018 – present



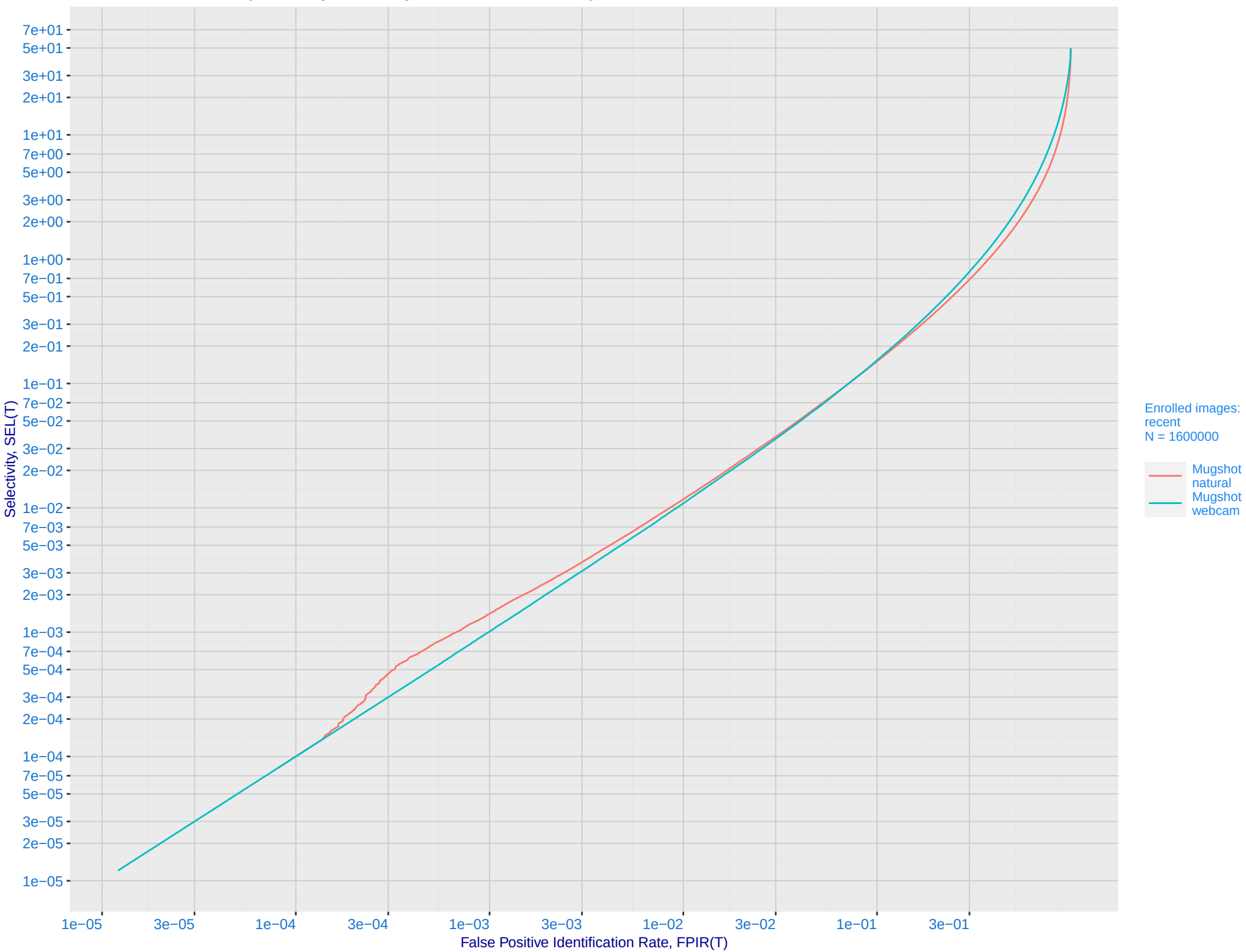
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals



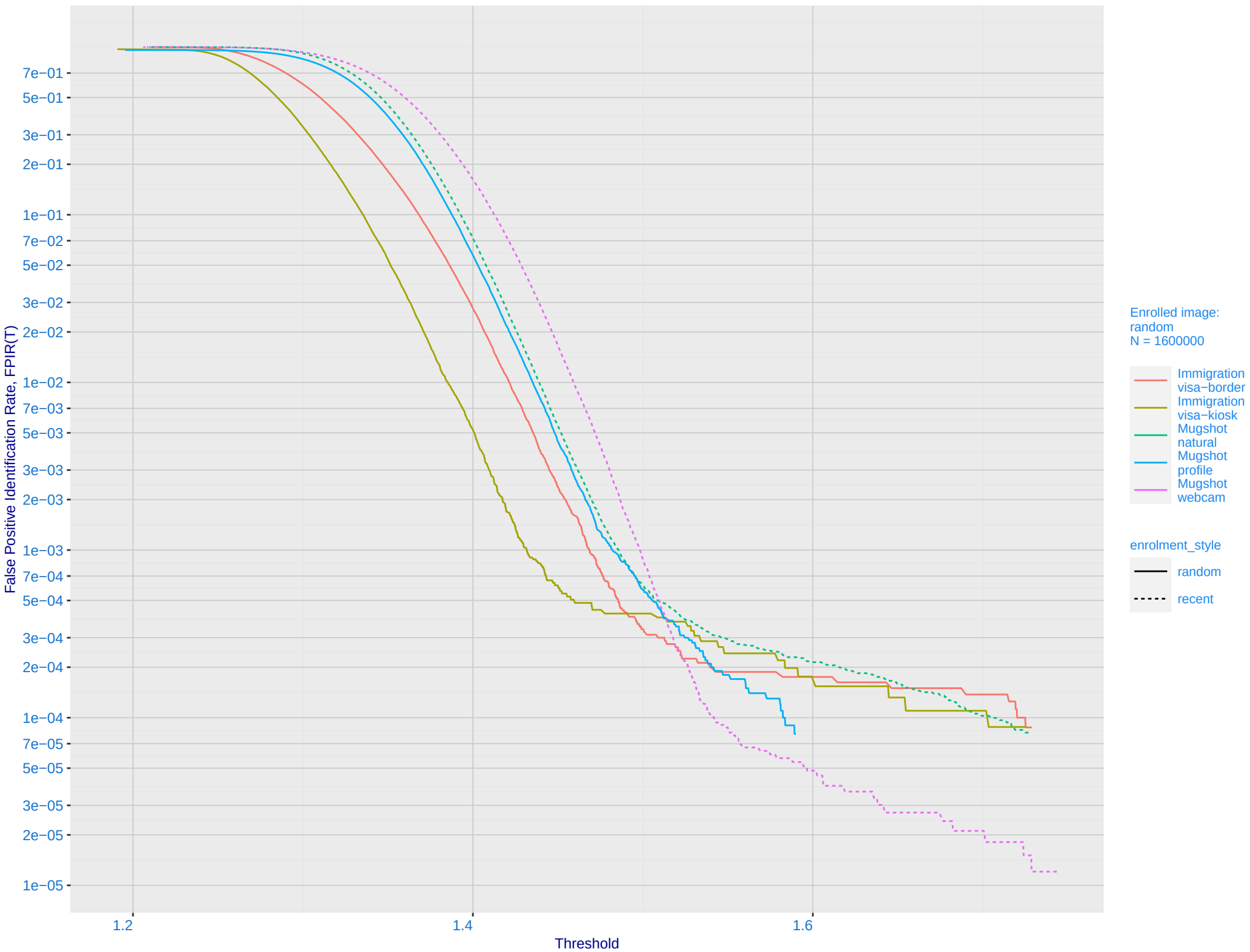
E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images



F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate

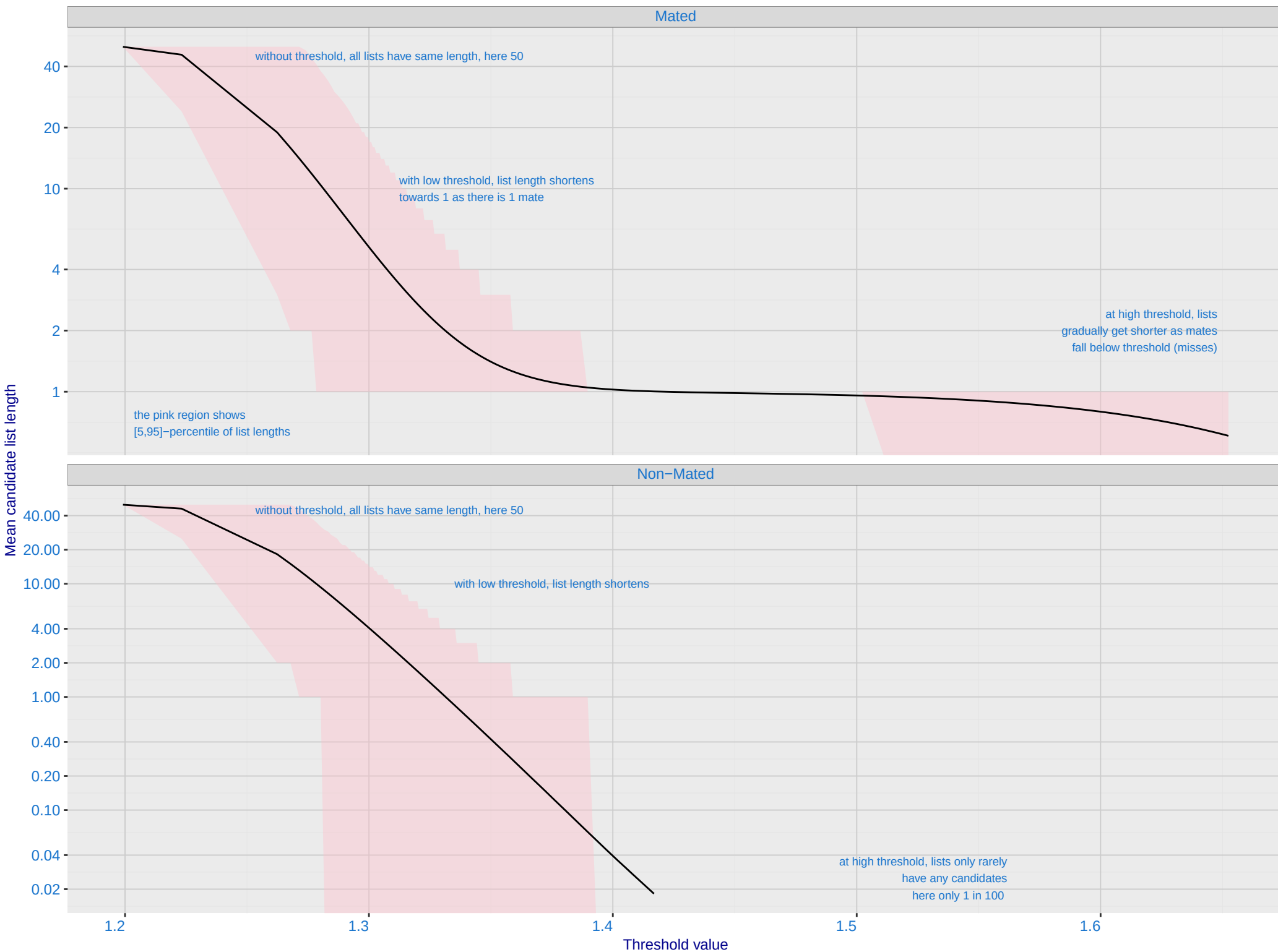


G: FPIR dependence on T by probe type for N = 1600000 subjects



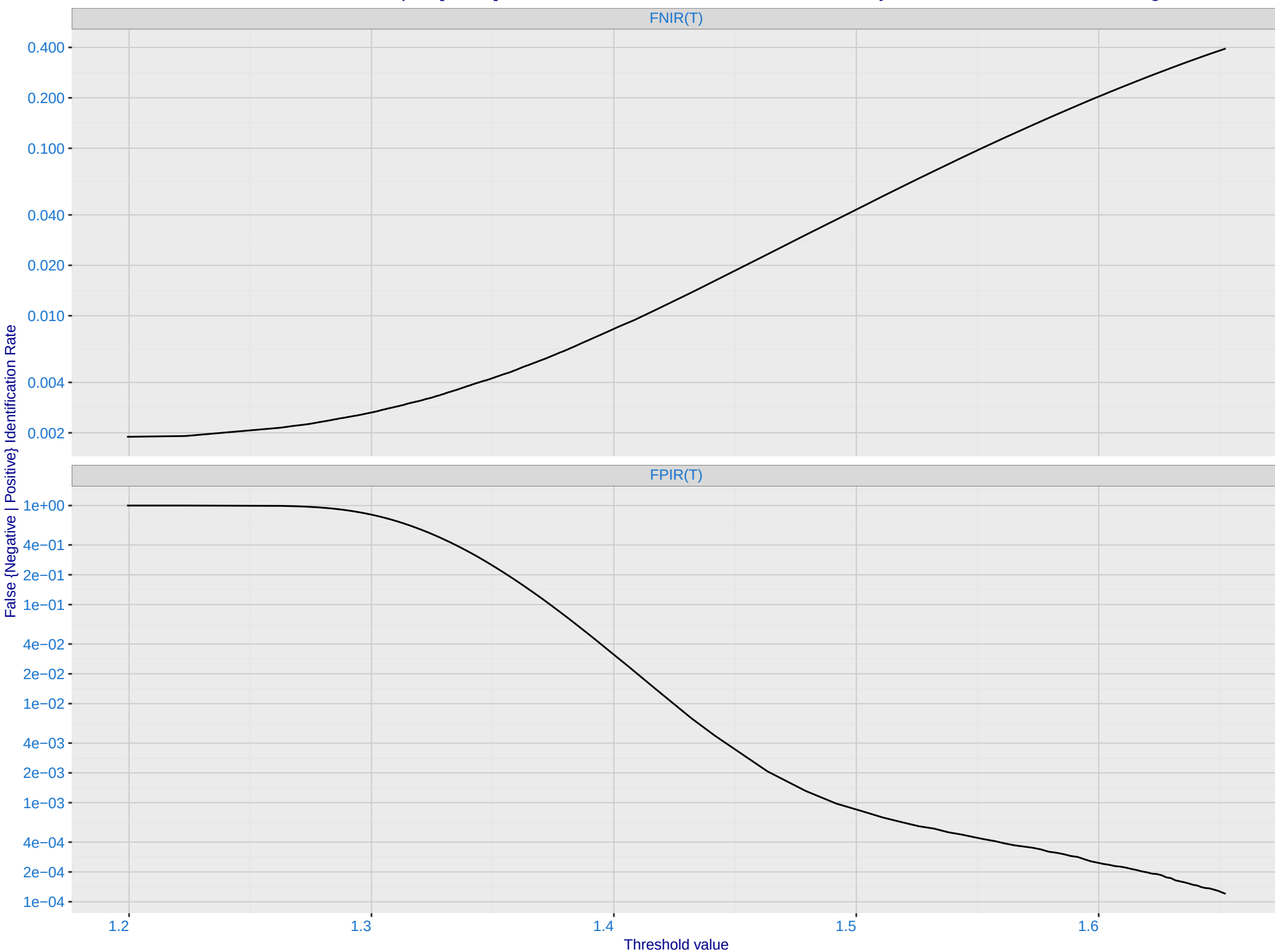
H: Reduced length candidate lists for human review

Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10–15 years later than enrollment image

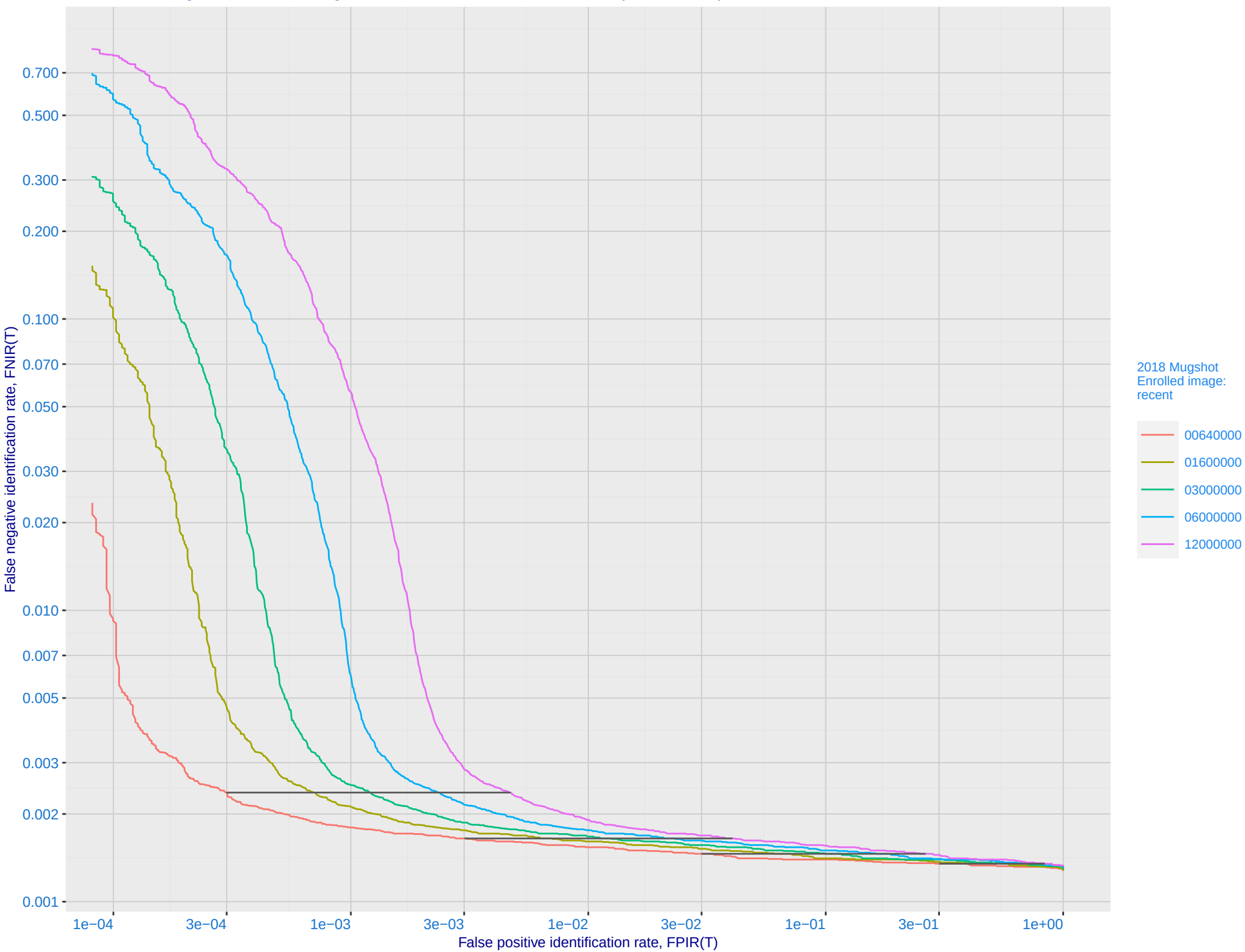


I: FNIR and FPIR dependence on threshold

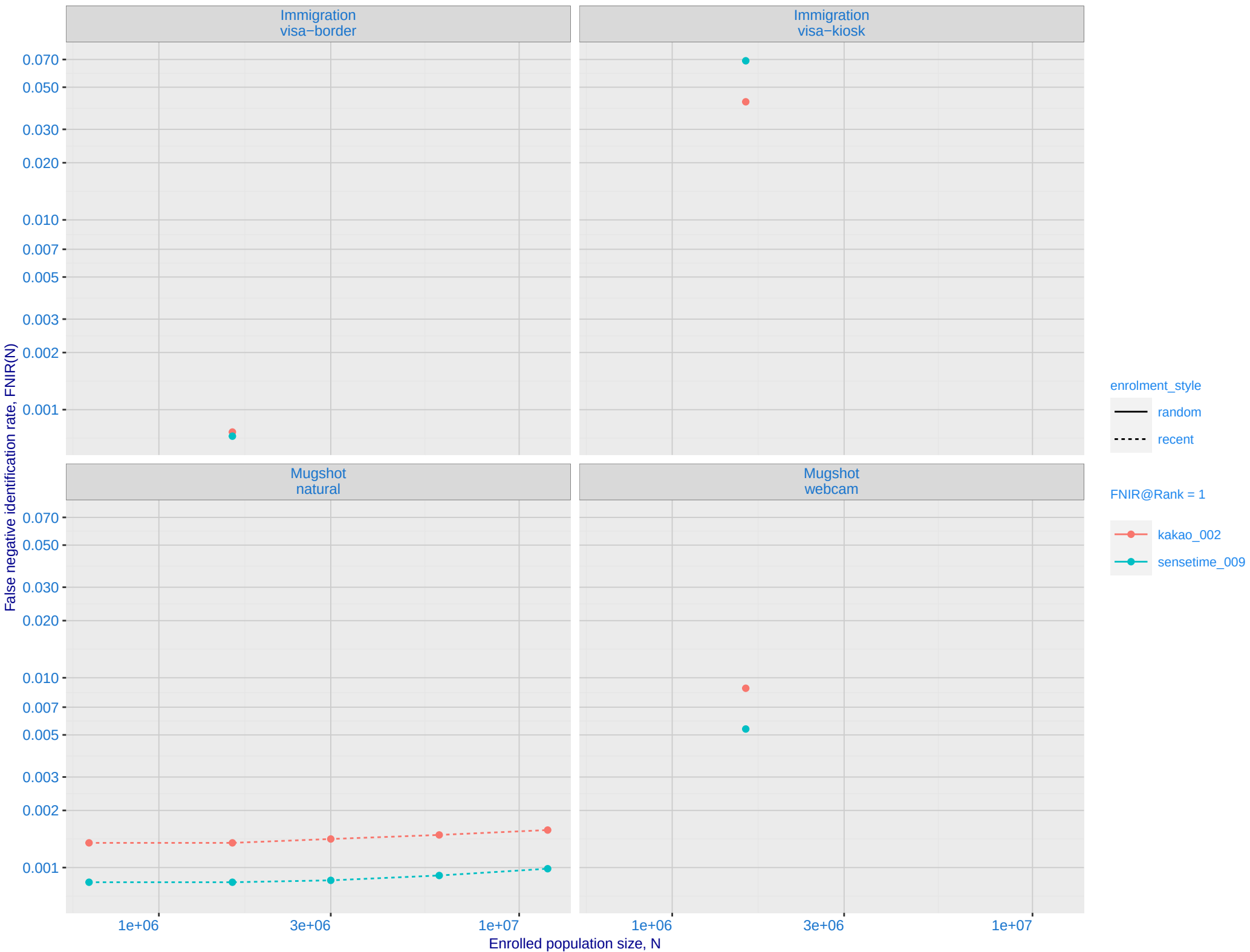
Dataset is border-border with time-lapse [10,15] YRS with N = 1600000. Probes are 10-15 years later than enrollment image



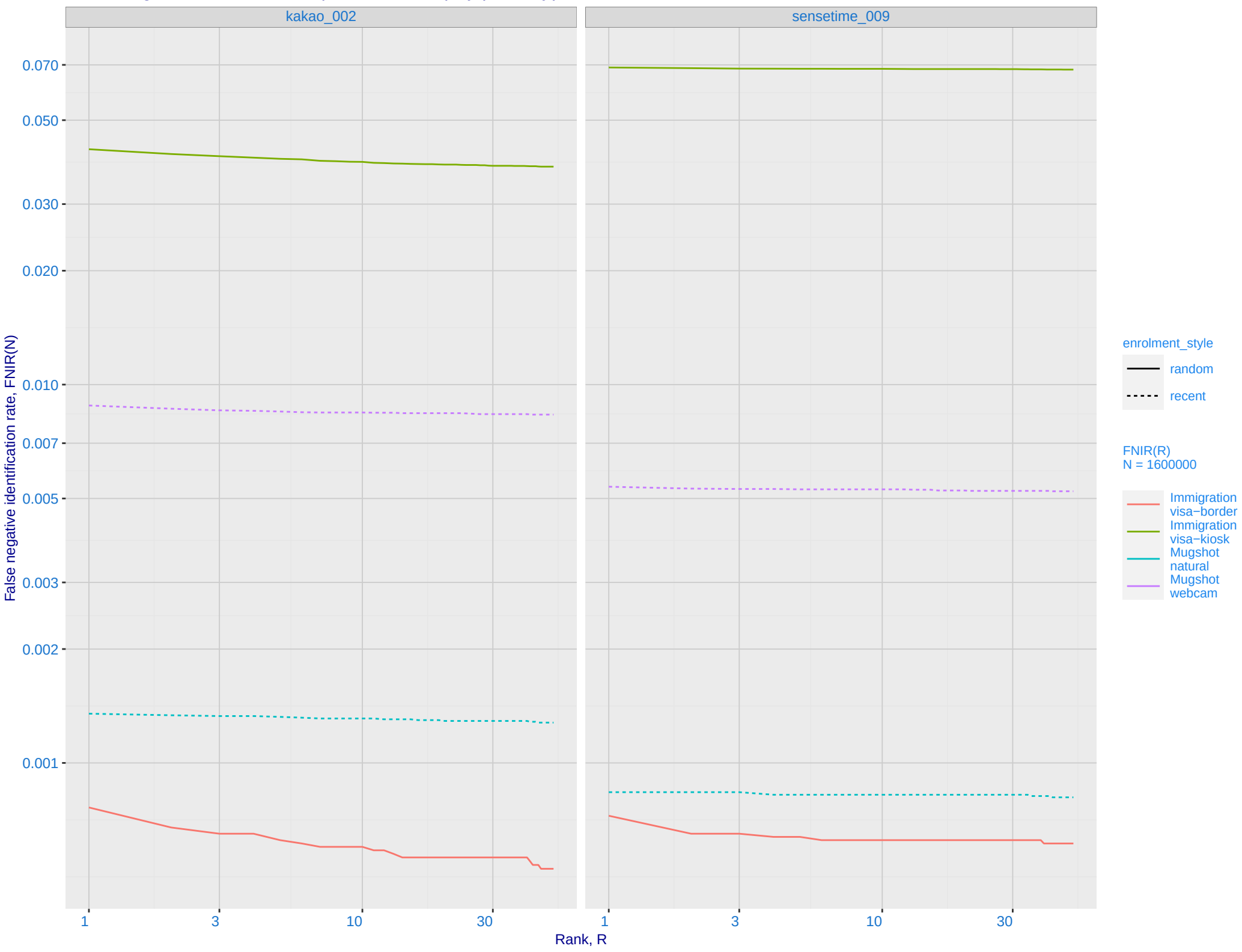
J: DET for Mugshot natural images and various N. Links connect points of equal threshold.



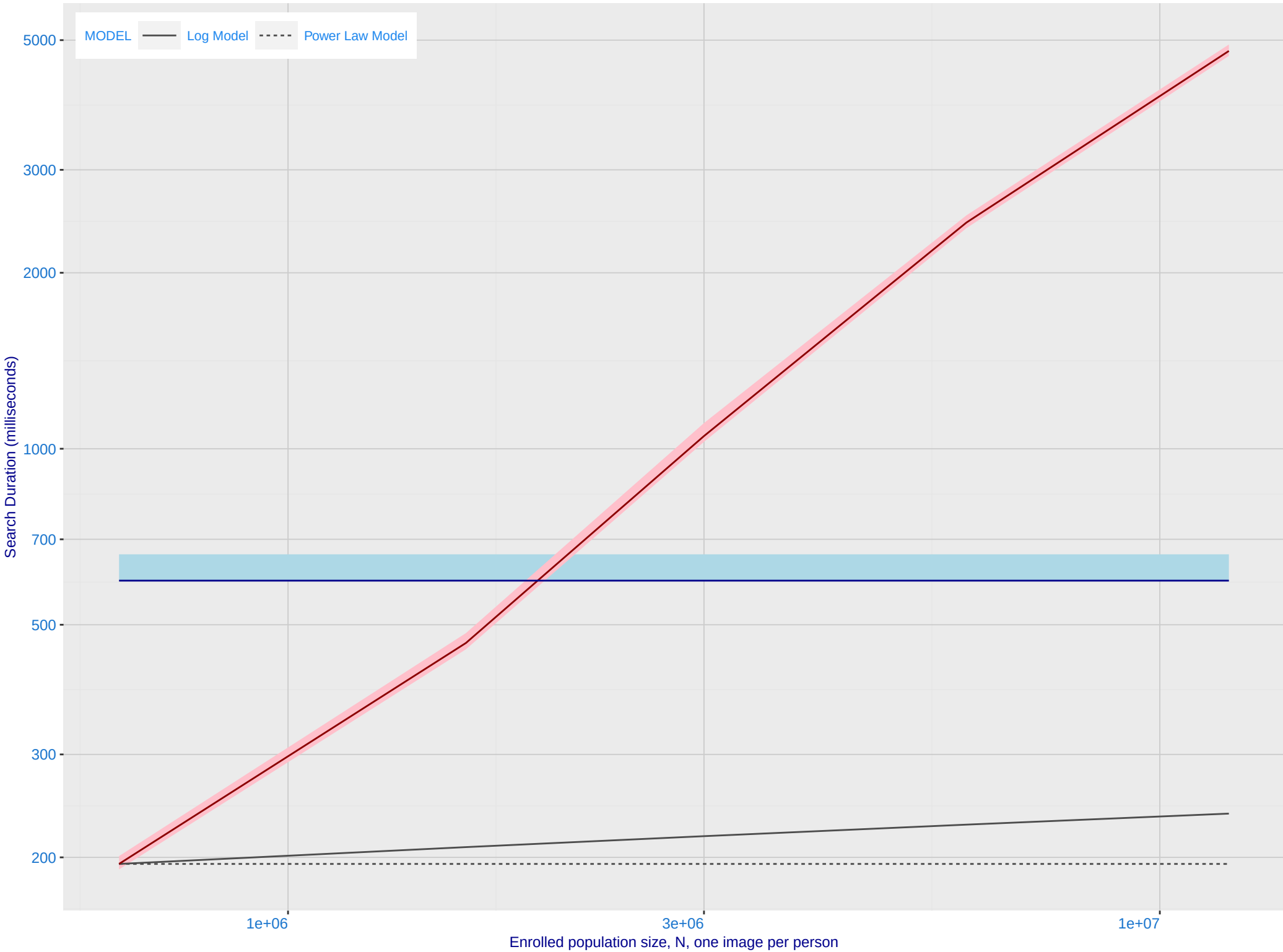
K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)



L: Investigational mode: FNIR(1600000, R, 0) by probe type



M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements. The template generation time is independent of N. The log and power-law models are fit to the first two (N,T) observations



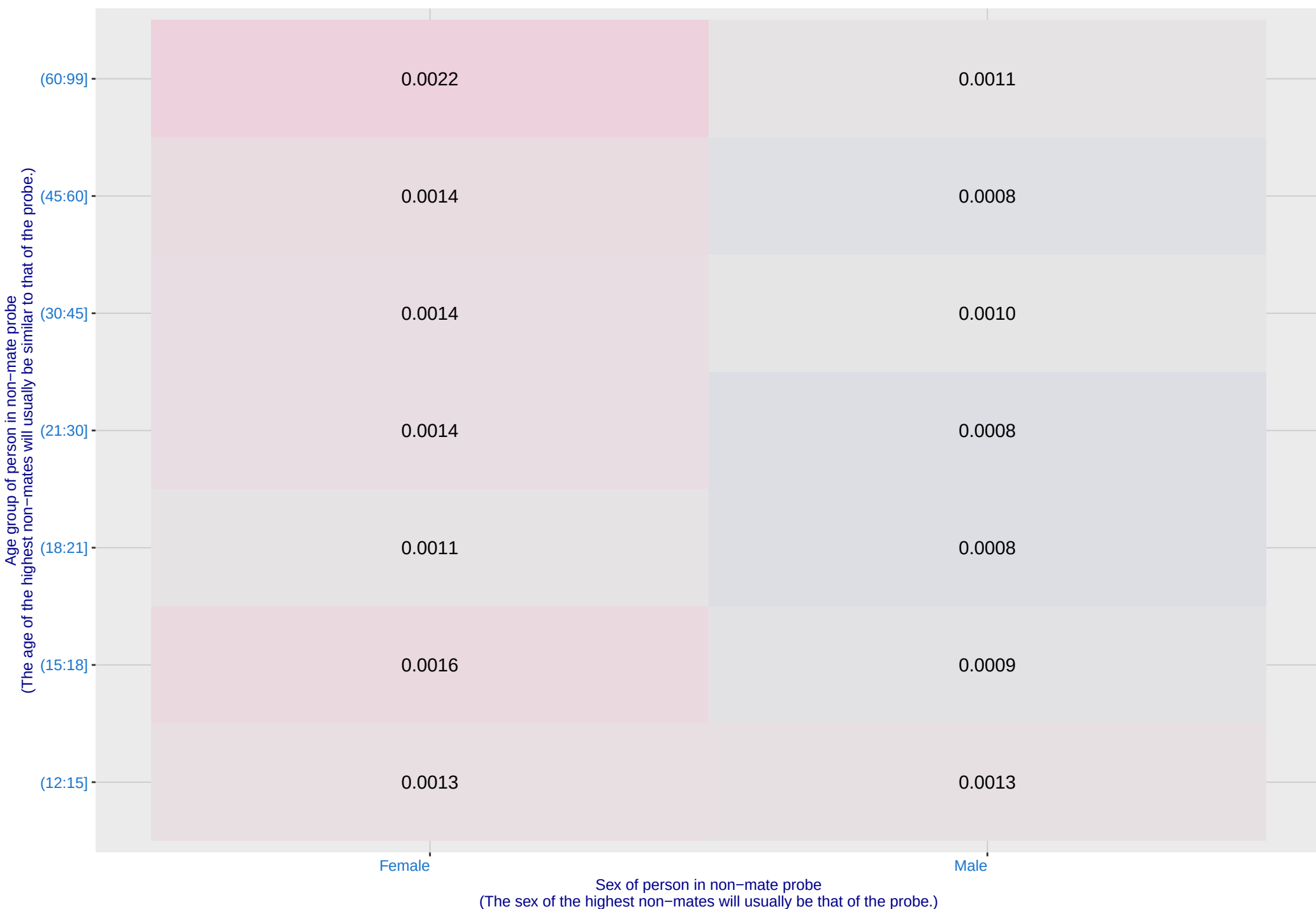
O: FNIR(T, N = 1.6 million) by sex, age and time-lapse. The top row gives investigational rank-1 miss rates. The bottom panels give high threshold for more lights-out identification with low FPIR.



P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.

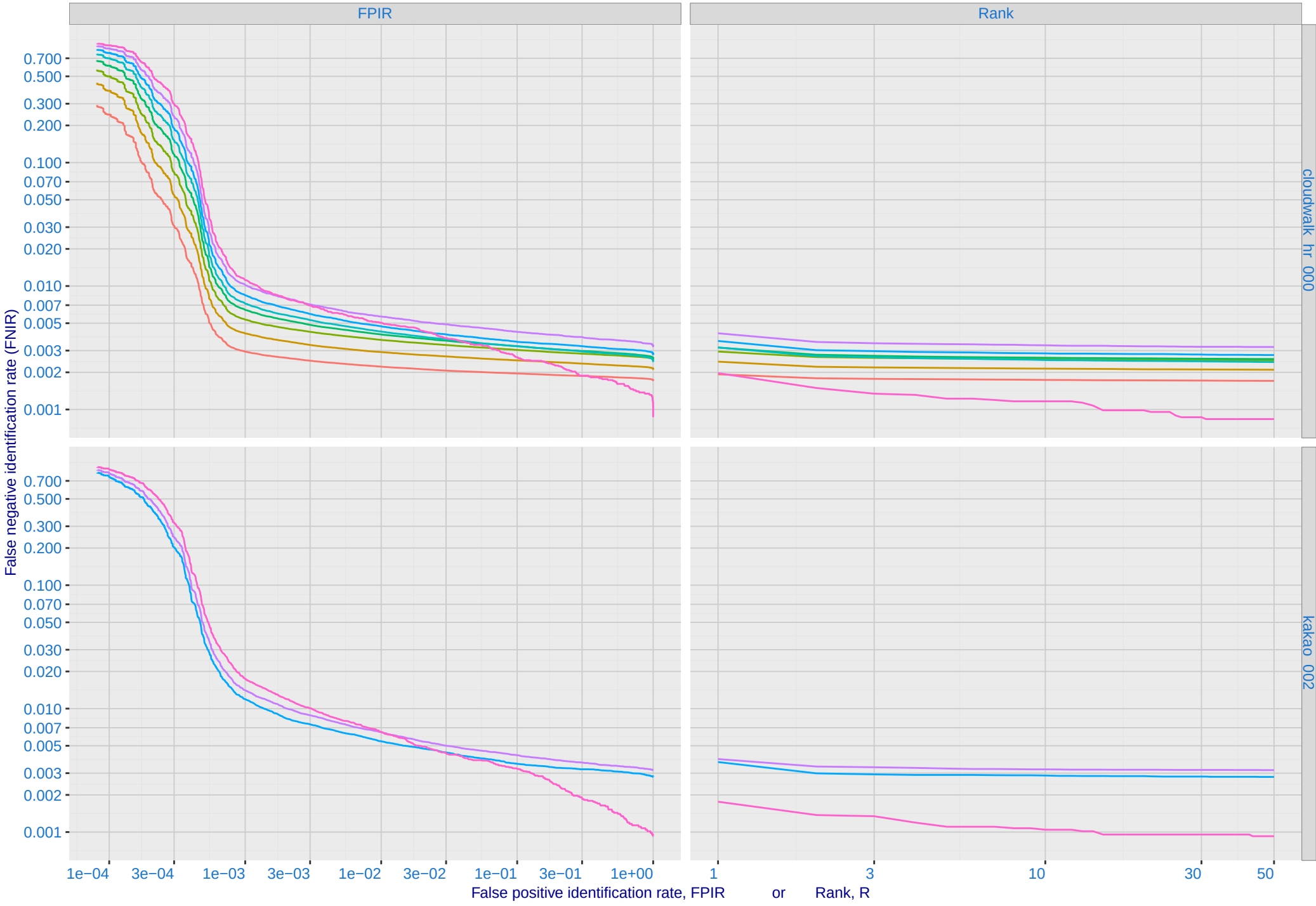
Algorithm: kakao_002, Dataset: Border-Crossing Ageing
Threshold: 1.482255 set to achive FPIR(30-45, Male) = 0.001

Color encodes log(FPIR)



Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing

Dataset: 2018 Mugshot N = 3068801



R: Decline of genuine scores with ageing, with some eventually dropping below typical thresholds shown by the horizontal lines

